

Sean McLean

**ALY 6050 – Enterprise to
Business Analytics**

Module 6 – Final Project

**2024 Winter Semester – Session
2**

Instructor: Marc Ellis

**Project: Two Optimization
Problems: (i) A Transshipment
Problem (ii) A Risk Minimizing
Problem**

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Introduction

The final project tackles two optimization problems that focus on concepts learned from the last module of the course. The first part uses a transshipment model with the goal of finding the shortest routes possible between six waste plants and three waste proposal sites that will lead to lower company costs. The second part incorporates a quadratic programming model to answer a risk minimizing problem that will provide better returns on investments for an individual's portfolio. An excel spreadsheet is used to generate equations for both parts of the project with Excel Solver being pivotal in the analysis of the problems.

Analysis

Part 1: Rockhill Shipping & Transport Company

Allen, who manages the South-Atlantic office of Rockhill Shipping & Transport Company, is in negotiations for a new shipping contract with Chimotoxic, a chemical manufacturing company. Chimotoxic requests Rockhill to transport waste products from its six plants to three waste disposal sites. Allen is apprehensive about this proposal due to potential hazards associated with transporting chemical waste, including risks to human health and the environment. Moreover, certain communities along the transportation routes may prohibit the shipment of hazardous materials, necessitating careful handling and reduced speeds. Allen has assessed the shipping costs for transporting waste barrels from each of the six plants to each of the three disposal sites, outlined in the following table:

Costs (\$)	<u>Waste Proposal Site</u>					
<u>Plant:</u>	<i>Orangeburg (P)</i>	<i>Florence (Q)</i>	<i>Macon (R)</i>		<u>Plant:</u>	<u>Waste per Week (bbl)</u>
Denver (A)	12	15	17		Denver	45
Morganton (B)	14	9	10		Morganton	26
Morrisville (C)	13	20	11		Morrisville	42
Pineville (D)	17	16	19		Pineville	53
Rockhill (E)	7	14	12		Rockhill	29
Statesville (F)	22	16	18		Statesville	38

Table 1: Shipping costs, per barrel of waste from six plants to three waste disposal sites

The six plants including Rockhill where the proposal is based out of is Denver, Morganton, Morrisville, Pineville, and Statesville, and they are designated in the left column of the above table. The three plants that the waste is transported to are Orangeburg, Florence, and Macon. The left table shows the transportation costs for each of the plants and waste sites and the right table displays the amount of waste that each plant compiles per week which are measured in individual barrels. In addition, the waste disposal sites in Orangeburg, Florence, and Macon have weekly maximum capacities of 65, 80, and 105 barrels, respectively. Each plant and site contain a letter metric used for the equations created in Excel for solving the questions being evaluated.

The objective function for minimizing the costs of transporting barrels to the nearest logical waste site is provided below:

$$Z = \sum C_{ij} X_{ij}, i=A,B,C,D,E,F; j= P,Q,R$$

In transportation or transshipment situations, the mission is to lower transportation expenses as much as possible with the consideration of the unit costs involved. We incorporate two indices (i and j) to manage the transportation logistics. The decision variables, shown in the objective function as X_{ij} , represent the quantity of units transported from source i to destination j. The first index represents the source, while the second index indicates the destination. The objective function X_{ij} is then the units moved from each specific source to every designated destination. As a result, the model involves the utilization of two indices, as shown by sources A, B, C, and D, and destinations P, Q, and R (Module 6, 2024)

Between all the plants and possible destinations there are 12 decision variables, and the non-negativity is a constraint that guarantees that the transportation of goods from each source to every destination involves only non-negative quantities, and the numbering of both sources and destinations begins from zero or any positive integer. The constraints pertain to the number of barrels that can be shipped out to all three of the waste sites and are indicated as a greater than or equal symbol which indicates that it is at least that amount. The constraints that stem from the destination waste sites are the sum of all the plants equaling the maximum number of barrels that they can accept per week. The less than or equal symbol indicates that the number of barrels they can take weekly can't exceed that amount (Evans, 2012).

<u>Transportation Linear Programming Example</u>	
Minimize: $Z = \sum C_{ij} X_{ij}, i=A,B,C,D,E,F; j= P,Q,R$	
Subject To (Constraints):	
Source A: $X_{ap}+X_{aq}+X_{ar} \geq 45$	
Source B: $X_{bp}+X_{bq}+X_{br} \geq 26$	
Source C: $X_{cp}+X_{cq}+X_{cr} \geq 42$	
Source D: $X_{dp}+X_{dq}+X_{dr} \geq 53$	
Source E: $X_{ep}+X_{eq}+X_{er} \geq 29$	
Source F: $X_{fp}+X_{fq}+X_{fr} \geq 38$	
Destination P: $X_{ap}+X_{bp}+X_{cp}+X_{dp}+X_{ep}+X_{fp} \leq 65$	
Destination Q: $X_{aq}+X_{bq}+X_{cq}+X_{dq}+X_{eq}+X_{fq} \leq 80$	
Destination R: $X_{ar}+X_{br}+X_{cr}+X_{dr}+X_{er}+X_{fr} \leq 105$	
Non-Negativity: $X_{ij} \geq 1$	

The table shown below was designed to show the relationship between the two variables and using Solver the best routes are created. For the Orangeburg site, only two plants are optimal

in transporting barrels to their site, with Statesville having the most barrels with 38. The Florence site has only three plants that should transport barrels there with the Pineville plant having the most shipments with 53. The Macon site has three optimal plants for barrel transport with the Denver plant having the most shipments at 45. This calculation by Solver is factoring in all constraints for all the sources involved and the costs associated with each transportation route per plant. The total cost for the most optimal solution regarding the best possible routes for minimizing costs for Allen at Rockhill is \$3613 per week. There are 233 barrels shipped weekly combined from all six plants and the maximum number that can be accepted from all three waste sites combined equals 250. Because the number barrels do not exceed the maximum total of barrels that the waste sites can take in weekly, there is some flexibility with where the transport can go while still factoring in that the cost still needs to be as minimal as possible.

The safety concerns with shipping hazardous waste while addressing community issues also needs to be addressed and handled properly when deciding on the most optimal routes. Because we are unaware of the distances between the plants and waste sites, some of the sources and destination cities and towns could be more suitable than other stops but might be too much of a risk when looking at road conditions and the health risks involved in certain communities.

	<u>Destination</u>					
<u>Plant:</u>	<i>Orangeburg (P)</i>	<i>Florence (Q)</i>	<i>Macon (R)</i>	Shipped From:		Total Cost Z
Denver (A)	0	0	45	45 ≥	45	3613
Morganton (B)	0	0	26	26 ≥	26	
Morrisville (C)	0	25	17	42 ≥	42	
Pineville (D)	0	53	0	53 ≥	53	
Rockhill (E)	27	2	0	29 ≥	29	
Statesville (F)	38	0	0	38 ≥	38	
Shipped To:	65	80	88		233	
	≤	≤	≤			
Demand:	65	80	105	250		

The second portion of the first question centers on whether to direct ship from each of the six plants to one of the three waste disposal sites. Allen is considering utilizing both the plants and disposal sites as intermediate shipping hubs. In this setup, trucks could unload at either a plant or a disposal site, with another truck transporting the load to the destination. Additionally, Rockhill would not incur any handling costs, as Chimitoxic has agreed to cover all expenses at both the plants and disposal sites. Essentially, Rockhill's only expense would be the transportation cost. Therefore, Allen seeks to investigate potential cost savings by dropping off and picking up loads at intermediate points rather than shipping them directly (Module 6, 2024).

Allen has totaled the costs of each plant and waste site if they were used as an intermediate shipping stop. The tables below provide these costs if the barrels were first transported to another plant and if the barrels were sent directly to a waste site before being sent to the destination which is another waste site in this scenario. The costs of intermediate transportation at first

glance indicates that it would most likely be cheaper to transport to another plant than another waste site as an intermediate hub.

	<u>Plant</u>					
<u>Plant:</u>	Denver	Morganton	Morrisville	Pineville	Rockhill	Statesville
Denver	\$---	\$3	\$4	\$9	\$5	\$4
Morganton	6	---	7	6	9	4
Morrisville	5	7	---	3	4	9
Pineville	5	4	3	---	3	11
Rockhill	5	9	5	3	---	14
Statesville	4	7	11	12	8	---

Table 2: Shipping costs, per barrel of waste from each plant to another plant

	<u>Waste Proposal Site</u>		
<u>Waste Disposal Site:</u>	Orangeburg	Florence	Macon
Orangeburg	\$---	\$12	\$10
Florence	12	---	15
Macon	10	15	---

Table 3: Shipping costs, per barrel of waste between the three waste disposal sites

The linear programming model for the revised transportation proposal has an adjusted objective function that adds the intermediate stops that are called nodes in this model. The nodes are labeled with the letters of the source and destination sites followed by the number one:

$$\text{Minimize: } Z = \sum C_{ij} X_{ij}, i=A,B,C,D,E,F; j=A1, B1, C1, D1, E1, F1, P1, Q1, R1, P,Q,R$$

The constraints of all the nodes are included in the linear programming model and are set up so that all the barrels that go through the nodes or intermediate sites must be the same number of barrels shipped out from that site. The conservation of flow principle governs intermediate points in transshipment problems, where no storage capacity exists. Incoming shipments must be promptly dispatched out, aligning with this fundamental principle (Module 6, 2024).

Transshipment Linear Programming Example	
Minimize: $Z = \sum C_{ij} X_{ij}, i=A,B,C,D,E,F; j=A1,B1,C1,D1,E1,F1,P1,Q1,R1,P,Q,R$	
Subject To (Constraints):	
Source A: $X_{Ap} + X_{Aq} + X_{Ar} \geq 45$	
Source B: $X_{Bp} + X_{Bq} + X_{Br} \geq 26$	
Source C: $X_{Cp} + X_{Cq} + X_{Cr} \geq 42$	
Source D: $X_{Dp} + X_{Dq} + X_{Dr} \geq 53$	
Source E: $X_{Ep} + X_{Eq} + X_{Er} \geq 29$	
Source F: $X_{Fp} + X_{Fq} + X_{Fr} \geq 38$	
Destination P: $X_{Ap} + X_{Bp} + X_{Cp} + X_{Dp} + X_{Ep} + X_{Fp} \leq 65$	
Destination Q: $X_{Aq} + X_{Bq} + X_{Cq} + X_{Dq} + X_{Eq} + X_{Fq} \leq 80$	
Destination R: $X_{Ar} + X_{Br} + X_{Cr} + X_{Dr} + X_{Er} + X_{Fr} \leq 105$	
Node A: $X_{Ab1} + X_{Ac1} + X_{Ad1} + X_{Ae1} + X_{Af1} + X_{Ap1} + X_{Aq1} + X_{Ar1} = X_{Ap} + X_{Aq} + X_{Ar}$	
Node B: $X_{Bb1} + X_{Bc1} + X_{Bd1} + X_{Be1} + X_{Bf1} + X_{Bp1} + X_{Bq1} + X_{Br1} = X_{Bp} + X_{Bq} + X_{Br}$	
Node C: $X_{Cb1} + X_{Cb1} + X_{Cd1} + X_{Ce1} + X_{Cf1} + X_{Cp1} + X_{Cq1} + X_{Cr1} = X_{Cp} + X_{Cq} + X_{Cr}$	
Node D: $X_{Db1} + X_{Db1} + X_{Dc1} + X_{De1} + X_{Df1} + X_{Dp1} + X_{Dq1} + X_{Dr1} = X_{Dp} + X_{Dq} + X_{Dr}$	
Node E: $X_{Eb1} + X_{Eb1} + X_{Ec1} + X_{Ed1} + X_{Ef1} + X_{Ep1} + X_{Eq1} + X_{Er1} = X_{Ep} + X_{Eq} + X_{Er}$	
Node F: $X_{Fb1} + X_{Fb1} + X_{Fc1} + X_{Fd1} + X_{Fe1} + X_{Fp1} + X_{Fq1} + X_{Fr1} = X_{Fp} + X_{Fq} + X_{Fr}$	
Node P1: $X_{Ap1} + X_{Bp1} + X_{Cp1} + X_{Dp1} + X_{Ep1} + X_{Fp1} = X_{Ap} + X_{Bp} + X_{Cp} + X_{Dp} + X_{Ep} + X_{Fp}$	
Node Q1: $X_{Aq1} + X_{Bq1} + X_{Cq1} + X_{Dq1} + X_{Eq1} + X_{Fq1} = X_{Aq} + X_{Bq} + X_{Cq} + X_{Dq} + X_{Eq} + X_{Fq}$	
Node R1: $X_{Ar1} + X_{Br1} + X_{Cr1} + X_{Dr1} + X_{Er1} + X_{Fr1} = X_{Ar} + X_{Br} + X_{Cr} + X_{Dr} + X_{Er} + X_{Fr}$	
Non-Negativity: $X_{ij} \geq 0$	

A new table was retooled to include the three waste sites since they are akin to the six plants in that they are also intermediate transport stops. Using Solver again in Excel to find the most optimal routes if the nodes are included in this plan, the final calculations show that each of the six plants are an intermediate transport stop for only one other plant. The three waste sites were all left out of this equation, indicating that they should not be used as an intermediate transport hub.

The total cost when using a transport node would be \$763 per week which would cut the costs almost in half if you are shipping directly to the waste site. An assessment of these results would be that there is one stop per plant that would be most ideal when using this model and would possibly suggest that it is the safest, the cheapest, and the shortest route in terms of mileage and time. The notion that the three waste sites were too expensive to be used as a connecting waste site looks as though it might be true considering the final calculations.

Costs (\$)	Waste Proposal Site			Plant						Total Cost
	Orangeburg (P)	Florence (Q)	Macon (R)	Denver	Morganton	Morrisville	Pineville	Rockhill	Statesville	
Denver (A)	12	15	17	\$1,000	\$3	\$4	\$9	\$5	\$4	763
Morganton (B)	14	9	10	6	1000	7	6	9	4	
Morrisville (C)	13	20	11	5	7	1000	3	4	9	
Pineville (D)	17	16	19	5	4	3	1000	3	11	
Rockhill (E)	7	14	12	5	9	5	3	1000	14	
Statesville (F)	22	16	18	4	7	11	12	8	1000	
Orangeburg	\$1,000	\$12	\$10	12	14	13	17	7	22	
Florence	12	1000	15	15	9	20	16	14	16	
Macon	10	15	1000	17	10	11	19	12	18	

Costs (\$)	Waste Proposal Site			Plant						Shipped From:	Supply
	Orangeburg (P)	Florence (Q)	Macon (R)	Denver	Morganton	Morrisville	Pineville	Rockhill	Statesville		
Denver (A)	0	0	0	\$0	\$45	\$0	\$0	\$0	\$0	45 =	45
Morganton (B)	0	0	0	0	0	0	0	0	26	26 =	26
Morrisville (C)	0	0	0	0	0	0	42	0	0	42 =	42
Pineville (D)	0	0	0	0	0	53	0	0	0	53 =	53
Rockhill (E)	0	0	0	0	0	0	29	0	0	29 =	29
Statesville (F)	0	0	0	38	0	0	0	0	0	38 =	38
Orangeburg	\$0	\$0	\$0	0	0	0	0	0	0	0	
Florence	0	0	0	0	0	0	0	0	0	0	
Macon	0	0	0	0	0	0	0	0	0	0	
Sum:	0	0	0	38	45	53	71	0	26		
Condition:	≤	≤	≤								
Demand:	65	80	105								

Part 2: Investment Allocations

A portfolio investor has chosen the following types of assets for their investment. The expected return for each asset type has been calculated using historical data that is presented in the table below:

	Expected Returns
Bonds	7%
High tech stocks	12%
Foreign stocks	11%
Call options	14%
Put options	14%
Gold	9%

The following table illustrates the covariance matrix of asset returns. On the diagonal, you'll find the variance of each individual asset, while off-diagonal entries indicate the covariances between pairs of assets:

	Bonds	High tech stocks	Foreign stocks	Call options	Put options	Gold
Bonds	0.001	0.0003	-0.0003	0.00035	-0.00035	0.0004
High tech stocks	0.0003	0.009	0.0004	0.0016	-0.0016	0.0006
Foreign stocks	-0.0003	0.0004	0.008	0.0015	-0.0055	-0.0007
Call options	0.00035	0.0016	0.0015	0.012	-0.0005	0.0008
Put options	-0.00035	-0.0016	-0.0055	-0.0005	0.012	-0.0008
Gold	0.0004	0.0006	-0.0007	0.0008	-0.0008	0.005

- (i) Suppose that our investor wishes to invest \$10,000 in this portfolio. Determine how he should allocate this investment to the individual assets in his portfolio in order to have a minimum baseline expected return of 11%, and at the same time, at a minimum risk.

The first steps when dissecting this problem is to create a quadratic programming model which will provide the objective function, the decision variables, and the constraints that are attached to the function. These are computed in the table shown below:

Decision Variables
x1: Fraction invested in bonds
x2: Fraction invested in high tech stocks
x3: Fraction invested in foreign stocks
x4: Fraction invested in call options
x5: Fraction invested in put options
x6: Fraction invested in gold
Objection Function
To Minimize: $\sigma^2 = x^T \Sigma x$
Expected Portfolio Return/Constraints
$x1+x2+x3+x4+x5+x6 = 10000$
$x_i \geq 0$
$0.07x1+0.12x2+0.11x3+0.14x4+0.14x5+0.09x6 \geq 0.11$
$u = 0.07x1+0.12x2+0.11x3+0.14x4+0.14x5+0.09x6$
$c^t=[0.07 \ 0.12 \ 0.11 \ 0.14 \ 0.14 \ 0.09]$
$u = c^t x$
Total Investment: $x_i \times R_i \geq \text{Minimum Return} \times 10000$
Baseline Expected Return: $u = \geq u_b$
$c^t x \geq u_b$
$\sum(x_i)=1$

The decision variables are the six assets in the investor's portfolio that include bonds, high tech stocks, foreign stocks, call options, put options, and gold. The objective function is to use an equation that will minimize the risk involved with investing \$10,000 into the assets in his portfolio. Mathematically, the total risk can be represented by the portfolio's variance. This variance, symbolized by σ^2 , is computed by initially pre-multiplying the covariance matrix by the transpose of X, followed by post-multiplying this outcome by the vector X. This value, or its square root σ , shows the overall risk that relates to the entire portfolio. The constraint $u=c^T x$ calculates the total expected return of the portfolio by multiplying the expected return of each asset by its corresponding allocation, adding up all six of the assets. It provides a quantitative measure of the overall expected return of the portfolio and is influenced by both the allocation process and the investors' expected returns of their assets. (Module 6, 2012). The constraints for the first part of the problem pertain to the expected portfolio returns and the risks of investing \$10,000 with an 11 percent baseline expected return. The sum of the expected returns of the assets must equal one or 100 percent of the investment as well, and there can't be any negative or nonlinear values from any of the decision variables (Evans, 2012).

The first step is to fill in the remaining part of the covariance matrix which will mirror the top right side, and for reference the expected returns of each asset are included in the tables as provided below:

	Expected Returns									
Bonds	7%	7	Bonds	0.001	0.0003	-0.0003	0.00035	-0.00035	0.0004	
High tech stocks	12%	12	High tech stocks	0.0003	0.009	0.0004	0.0016	-0.0016	0.0006	
Foreign stocks	11%	11	Foreign stocks	-0.0003	0.0004	0.008	0.0015	-0.00055	-0.0007	
Call options	14%	14	Call options	0.00035	0.0016	0.0015	0.012	-0.0005	0.0008	
Put options	14%	14	Put options	-0.00035	-0.0016	-0.00055	-0.0005	0.012	-0.0008	
Gold	9%	9	Gold	0.0004	0.0006	-0.0007	0.0008	-0.0008	0.005	
Target Return	11%		Expected Return	7%	12%	11%	14%	14%	9%	11%

With the expected returns of all the assets and the co-variance matrix filled in we can solve the first part of the problem by using Solver again. Factoring in these values with the objective function, the decision variables, and the constraints, the investment of \$10,000 is dispersed evenly across the six assets based off the inputs. The weights show the percentages of each asset with the bonds receiving most of the investment at 52 percent. The financial values were then calculated by taking the weights and multiplying them by 10,000. The bonds asset total was \$5182.20 while the call options asset did not receive any of the investment. With a minimum baseline expected return of 11 percent for this problem, the third column was calculated by taking the financial total of each asset from the previous column and multiplying that value by 11. With 11 percent of \$10,000 equaling \$1,100, the total of all six assets expected returns at 11 percent equaled just under \$1,100 (LondonPhD, 2016).

The calculations from Excel Solver also computed the values of the portfolio returns, portfolio variance and the standard deviation of the portfolio variance. The portfolio returns before considering the 11 percent expected return equaled 0.092 for the average expected returns of all six assets. The portfolio variance quantifies the extent to which returns can deviate from the average return. A smaller variance signifies more consistent returns and lower volatility. In this instance, a variance of 0.000461 indicates that portfolio returns exhibit relatively less fluctuation around the average return (Evans, 2012). This return value should instill confidence in the investor who has just invested a large sum of money into the assets of his portfolio.

The standard deviation represents the square root of the variance and serves as an indicator of the spread of returns from the average. It gauges the level of risk inherent in the portfolio being assessed in the problem. A smaller standard deviation indicates reduces the risk or volatility. In this scenario, a standard deviation of 0.021469 suggests that portfolio returns typically diverge by approximately 2.15% from the average return (Bluman, 2017).

	Weights	Weights & Investment	11% Expected Return	Portfolio Returns	0.092381
Bonds	0.52	5182.200008	570.0420008	Portfolio Variance	0.000461
High tech stocks	0.05	500	55	Portfolio SD	0.021468
Foreign stocks	0.19	1900	209		
Call options	0.00	0	0		
Put options	0.15	1500	165		
Gold	0.10	900	99		
			0		
	1.00	9982.20	1098.042001		

An analysis of the investment indicates that after investing \$10,000 in the portfolio the asset that should be allocated the most money to is the bonds asset followed by the foreign stocks asset and the put options asset. The only risky asset in the portfolio is the call options asset that received no money that was allocated to it. The lack of fluctuations in the values calculated from Solver in the portfolio variance and portfolio returns suggests there is minimal risk with making this investment.

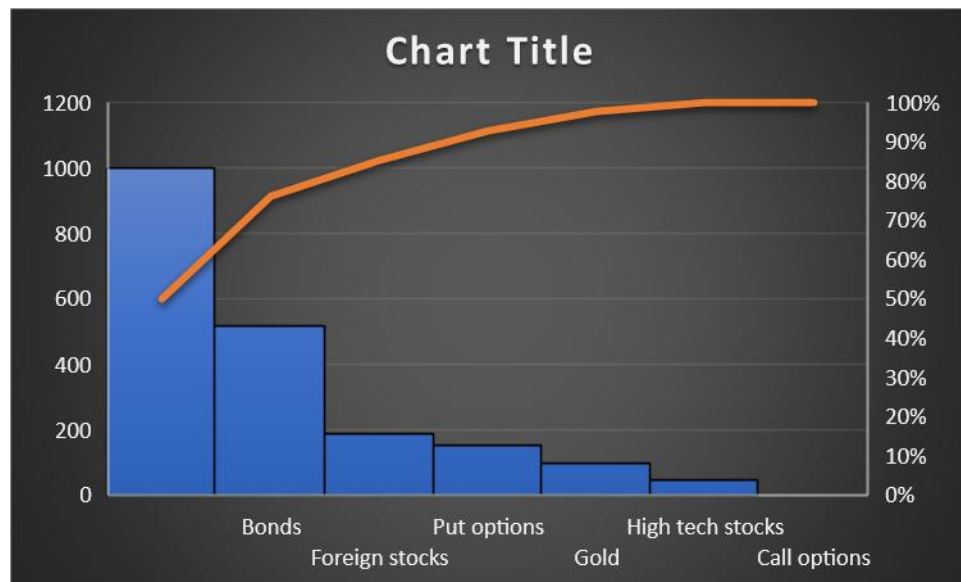
- (ii) Let the solution pair be denoted by (r, e), where “r” denotes the minimized risk and “e” denotes the expected portfolio return after the problem is solved. Use successive values of 10%, 10.5%, 11%, 11.5%, 12%, 12.5%, 13% and 13.5% as the baseline return values to obtain eight pairs of solutions (r, e). Plot “e” versus “r”. Explain whether there exists a pattern in this plot. In other words, explain, in your opinion, the type of mathematical relationship that “r” and “e” may have.

Using the previous calculations that used the investment and factored in the expected return value of 11 percent, the table of expected returns was formulated. This was completed by taking the weights and investment values of each asset from the investment of \$10,000 and multiplying those values by the expected return percentage that is shown in the table below. The range of expected returns shows the increases that were calculated based on the increase in expected returns. This would entail a certain amount of risk because the average portfolio returns of the six assets are just over nine percent. Despite the expected returns increasing in the table, there are several assets like high-tech stocks and call options that do not return enough financial value to consider raising the minimum baseline expected return.

	10% Expected Return	10.5 Expected Return	11% Expected Return	11.5% Expected Return	12% Expected Return	12.5 Expected Return	13% Expected Return	13.5% Expected Return
Bonds	518.2200008	544.1310008	570.0420008	595.9530009	621.8640009	647.775001	673.686001	699.597001
High tech stocks	46.24586469	48.55815792	50.87045116	53.18274439	55.49503763	57.80733086	60.1196241	62.4319173
Foreign stocks	187.6704536	197.0539763	206.4374989	215.8210216	225.2045443	234.588067	243.9715897	253.355112
Call options	0	0	0	0	0	0	0	0
Put options	152.0875074	159.6918827	167.2962581	174.9006335	182.5050088	190.1093842	197.7137596	205.318135
Gold	95.7751736	100.5639323	105.352691	110.1414496	114.9302083	119.718967	124.5077257	129.296484
	0	0	0	0	0	0	0	0
	999.999	1049.99895	1099.9989	1149.99885	1199.9988	1249.99875	1299.9987	1349.99865

A Pareto chart plot was created that is on the next page that shows the distribution of the numbers across a few of the categories that are being evaluated. The plot shows the relationship

between the assets expected returns and the total percentages of the returns. The first blue bar shows the total expected returns values with the assets following in order of rank in values. While these numbers are decreasing, the orange line is going in the other direction. This shows the cumulative effect of the total percentages which are increasing. The line starts at around 600, which looks like the average of the expected returns on the bonds asset and increases to 100 percent over the plot.



An evaluation of the mathematical relationship between the minimized risk(r) and expected portfolio return (e) shows mainly a pattern of a sharp increase in monetary returns that could be a risky choice in selecting a higher baseline return value. It would be dependent upon the investor to decide on what kind of risk they are willing to take when assessing their portfolio. I do think there is a strong connection between the two variables being looked at even though I am unsure if this formula is adequate for looking at this relationship. The question overall is a good indicator of how decision variables like covariance matrices can show relationships between variables and incorporating constraints to the Excel Solver for solving investments problems.

Conclusion

This final project demonstrated the lessons learned from this week's module by answering two questions that looked at how to optimize solutions. The transshipment model was a useful way to see how identifying the best transportation routes can be more efficient for the company and be beneficial in decreasing costs. This is also a very effective example when looking at supply and demand and how they can impact company profits. The second half of the project looked at investment allocations and how quadratic and nonlinear models can be implemented in evaluating financial concerns and strengthening portfolios. The importance of external factors

like risk analysis was used in configuring expected returns on investments. The usage of Excel Solver was a convenient tool in setting up and executing problems for this project and is highly recommended for future use. This was a fantastic way to wrap up the course and had its challenges and rewards at its conclusion.

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